

Homework #6

1. McIntyre Problem 5.17
2. McIntyre Problem 5.34
3. McIntyre Problem 6.2 part (i)
4. Complete the following worksheet-style questions about superposition.

Any superposition of two energy eigenfunctions can be written in the form

$$\psi(x) = e^{i\alpha} \left(c_n \varphi_n(x) + c_m e^{i\beta} \varphi_m(x) \right),$$

where

- $\varphi_i(x)$ is the i th energy eigenfunction;
- $n < m$;
- α and β are real numbers (possibly zero); and
- c_n and c_m are **positive** real numbers.

When written in this way, α is often called the **overall phase** of the wave function and β is called the **relative phase** between the energy eigenfunctions $\varphi_n(x)$ and $\varphi_m(x)$.

3. What is the **overall phase** of each of the wave functions $\psi_A(x)$, $\psi_B(x)$, $\psi_C(x)$, and $\psi_D(x)$?
4. Does the probability density depend on the **overall phase** of the wave function? Explain.

Describe how your answer is consistent with your response to question 2 on the previous page.

5. What is the **relative phase** between the energy eigenfunctions $\varphi_1(x)$ and $\varphi_2(x)$ in each of the wave functions $\psi_A(x)$, $\psi_B(x)$, $\psi_C(x)$, and $\psi_D(x)$?
6. Does the probability density depend on the **relative phase** between the energy eigenfunctions $\varphi_1(x)$ and $\varphi_2(x)$ in a given wave function? Explain.

Describe how your answer is consistent with your response to question 1 on the previous page.

Consider three pure ensembles— E , F , and G —of identical particles, each of which is in an identical infinite square well extending from $x = 0$ to $x = L$. The wave functions of the particles in the three ensembles are given below.

$$\psi_E(x) = \frac{1}{\sqrt{2}} e^{i\alpha_1} (\varphi_1(x) + e^{i\beta_1} \varphi_2(x))$$

$$\psi_F(x) = \frac{1}{\sqrt{2}} e^{i\alpha_2} (\varphi_1(x) + e^{i\beta_1} \varphi_2(x)) \qquad \psi_G(x) = \frac{1}{\sqrt{2}} e^{i\alpha_1} (\varphi_1(x) + e^{i\beta_2} \varphi_2(x))$$

Here, α_1 , α_2 , β_1 , and β_2 are real numbers with $\alpha_2 \neq \alpha_1$ and $\beta_2 \neq \beta_1$. Thus, the only difference between $\psi_E(x)$ and $\psi_F(x)$ is the overall phase and the only difference between $\psi_E(x)$ and $\psi_G(x)$ is the relative phase between $\varphi_1(x)$ and $\varphi_2(x)$.

7. Suppose you were to select a large number of particles at random from either **ensemble E** or **ensemble F** (not from both) and measure their positions. Would you be able to determine whether the particles came from ensemble E or ensemble F ? Explain.

8. Suppose you were to select a large number of particles at random from either **ensemble E** or **ensemble G** (not from both) and measure their positions. Would you be able to determine whether the particles came from ensemble E or ensemble G ? Explain.